

Tutorial 3: Symmetry

Homework: Exercises marked with red stars, with points adding to 10.

Affine Transformations

The informal definition of an affine transformation is: it maps straight objects (lines, planes, hyperplanes) to straight objects. Affine transformations F that preserve the origin (in other words: $F\mathbf{0} = \mathbf{0}$), are called **linear**. The formal mathematical definition is as follows:

Definition

A transformation on $\mathbb{R}^n$ is called **linear** if the following holds:

- $F(\mathbf{p} + \mathbf{q}) = F\mathbf{p} + F\mathbf{q}$ for every $\mathbf{p}$ and $\mathbf{q}$,
- $F(\alpha\mathbf{p}) = \alpha F\mathbf{p}$ for every $\mathbf{p}$ and every real number α .

Notice that this formal definition is something we can reasonably check for a specific F , compared with proving that any straight object remains straight. Why this definition implies that linear transformations indeed preserve straightness is rather technical, and we will not go into details here.

That linear transformations preserve the origin is easy, if you know the trick.

EXERCISE

- 1 Prove that if F is linear (according to the definition above), then $F\mathbf{0} = \mathbf{0}$.

Affine transformations also retain straightness, but they may map $\mathbf{0}$ to a point that is not the zero vector. Still, you can use linear maps to describe affine transformations:

Theorem

Every affine transformation F can be uniquely written as the composition of a linear transformation followed by a translation.

A translation is defined as a transformation that shifts every vector over a *fixed* distance and direction. More precise:

Definition

The **translation over $\mathbf{b}$** is the transformation $T\mathbf{p} = \mathbf{p} + \mathbf{b}$.

An alternative way to express the previous theorem is as follows:

Theorem

If F is an affine transformation on $\mathbb{R}^n$, then there exists a linear transformation A and a vector $\mathbf{b}$ such that

$$F\mathbf{p} = A\mathbf{p} + \mathbf{b} \quad \text{for all } \mathbf{p} \text{ in } \mathbb{R}^n.$$

The transformation A and vector $\mathbf{b}$ are uniquely determined.

Let's denote the translation over $\mathbf{b}$ as $T_{\mathbf{b}}$. The previous theorem says that we can write every affine transformation F as a composition: $F = T_{\mathbf{b}}A$ for some linear transformation A and some vector $\mathbf{b}$. We call the product $T_{\mathbf{b}}A$ a **decomposition** of F .

If an affine transformation F is given, finding A and $\mathbf{b}$ such that $F = T_{\mathbf{b}}A$ is not difficult. We start with $\mathbf{b}$. Observe that

$$F\mathbf{0} = T_{\mathbf{b}}A\mathbf{0} = A\mathbf{0} + \mathbf{b} = \mathbf{0} + \mathbf{b} = \mathbf{b},$$

in other words: $\mathbf{b}$ is always equal to $F\mathbf{0}$ (that is why $T_{\mathbf{b}}$ is unique).

In order to find A we need the inverse of $T_{\mathbf{b}}$.

EXERCISES

- 2 (a) Show that for every $\mathbf{b}$ and every $\mathbf{c}$ the following holds:

$$T_{\mathbf{b}}T_{\mathbf{c}} = T_{\mathbf{b}+\mathbf{c}}.$$

- (b) Show that the identity is a translation.
(c) Show that every translation $T_{\mathbf{b}}$ is invertible.

3 Assume $F = T_{\mathbf{b}}A$.

- (a) Express A as a composition of F and a translation.
- (b) Express $A\mathbf{p}$ as a formula with F , $\mathbf{p}$ and $\mathbf{0}$.

Isometries

Isometries are distance preserving functions:

Definition

An **isometry on** V is a transformation F of V that preserves distance:

$$\text{for all } \mathbf{p}, \mathbf{q} \in V: \text{dist}(\mathbf{p}, \mathbf{q}) = \text{dist}(F\mathbf{p}, F\mathbf{q}).$$

Distance preservation seems a quite innocuous condition, but it imposes severe restrictions on the transformation. For one, isometries are *invertible*. But there is more: isometries are composed of just a few elementary transformations:

Theorem

If F is an isometry of $\mathbb{R}^2$ or $\mathbb{R}^3$, then F is composed of one or more of the following transformations (which are isometries themselves):

- a translation,
- a rotation,
- a reflection.

The proof is quite technical and elaborate, and is left as an elective exercise (see exercise 15).

You can refine the theorem even further: you don't need arbitrary rotations, rotations about the origin will do. Same thing for the reflections: only reflections across a line or plane through the origin are needed.

For the remainder of this section we restrict ourselves to planar isometries. We define the following elementary transformations ($O = (0,0)$ is the origin):

Definition

- For $\theta \in \mathbb{R}$, we denote by R_θ the **rotation** around O over an angle θ with θ in Radians (remember that a full turn of 360° equals 2π Radians).
- For a line ℓ through O , we denote the **reflection** across ℓ is denoted by S_ℓ . It transforms each point $\mathbf{p}$ to another one at the same distance to ℓ but on the other side of it, along the line passing through $\mathbf{p}$ and intersecting ℓ at an angle of $\pi/2$ (90° degrees).

It is not difficult to see that these transformations are isometries. All isometries are affine. Not all isometries are linear, but R_θ and S_ℓ are. Therefore they can be represented by a matrix. To find the matrix of a linear transformation, we only have to calculate the images of the standard base vectors, and assemble them column-wise in a matrix. In $\mathbb{R}^2$, the base vectors are $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$.

EXERCISES

3

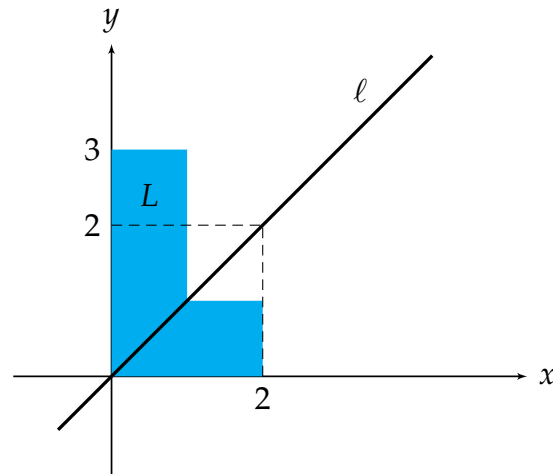
4 Find the matrix of the following linear transformations of the plane:

- The reflection S_ℓ with ℓ the line $y = -x$,
- The rotation $R_{\pi/3}$,
- The compositions $S_\ell R_{\pi/3}$ and $R_{\pi/3} S_\ell$.

3

5 A **glide reflection** is a combination of a translation and a reflection. If the reflection is about a line ℓ , then the translation must be parallel to ℓ . Let $\mathbf{b} = (2, 2)$ and let ℓ be the line $y = x$.

- If G is the glide reflection composed of $T_{\mathbf{b}}$ and S_ℓ , find the image under G of the L-shaped object L of Figure 1.
- Does it matter whether you write $G = T_{\mathbf{b}} S_\ell$ or $G = S_\ell T_{\mathbf{b}}$?
- Consider instead the vector $T_{\mathbf{a}} = (0, 2)$. Do we have $T_{\mathbf{a}} S_\ell = S_\ell T_{\mathbf{a}}$?

Figure 1: Region L .

- 6 Reflections and rotations do not have overlap: there doesn't exist a line ℓ and an angle θ such that $R_\theta = S_\ell$.
Prove this.

7 The order of isometries

Let F be an isometry. The n -th power of F is defined as the composition of n copies of F . If $n = 0$, then $F^0 = I$ by definition.

Take a look at the non-negative powers $I, F, F^2, F^3, \dots$. Either they are all different, or for some pair of non-negative integers m and k with $m \neq k$ we have $F^m = F^k$.

- (a) Assume that the latter is the case. Let n be the smallest positive integer for which F^n repeats, that is:
- (1) $I, F, F^2, F^3, \dots, F^{n-1}$ are all different, but
 - (2) $F^n \in \{I, F, F^2, F^3, \dots, F^{n-1}\}$.

Show that $F^n = I$.

The number n is called the **order** of F , and is denoted as $o(F)$.

If all non-negative powers $I, F, F^2, F^3, \dots$ are different we say that the order of a is infinite, denoted: $o(F) = \infty$.

- (b) Find all isometries with order 1.



8 Determine the order of the following isometries:

- (a) The translation $T_{\mathbf{b}}$ with $\mathbf{b} = (2, 0)$,
- (b) The rotation $R_{-\pi/5}$,
- (c) The rotation $R_{\sqrt{2}\pi}$,
- (d) The reflection S_{ℓ} with ℓ the Y -axis.

An important result in group theory is the following theorem:

Let G be a finite group. Then all elements a in G have finite order, and $o(a)$ is a divisor of $|G|$.

The number $|G|$ denotes the number of elements of G , and is called **order** of G .

9 Cyclic groups

Let F be an isometry of order n . We define the **cyclic group generated by F** (denoted as $\langle F \rangle$) as the set of all different powers F^k , so

$$\langle F \rangle = \{I, F, F^2, \dots, F^{n-1}\}.$$

- (a) Show that every composition of two elements in $\langle F \rangle$ always yields an element of $\langle F \rangle$.
- (b) Show that the inverse of every element in $\langle F \rangle$ is also an element of $\langle F \rangle$.

Symmetry

EXERCISES

10 The regular n -gon, denoted by $\{n\}$ (see also the tutorial of week 1), shows some symmetry. Indeed, a rotation about its center with rotation angle $\frac{2\pi}{n}$ leaves the n -gon invariant.

- (a) Let ℓ be a line through the midpoint of the n -gon. What are the requirements for ℓ such that a reflection across ℓ leaves $\{n\}$ invariant?

- (b) Find the total collection of rotations and reflections that leave $\{n\}$ invariant.

The symmetry group of $\{n\}$ is called the dihedral group, which we denote as D_n . It can be characterized as any group of isometries of $\mathbb{R}^2$ with the following properties:

1. It contains a rotation R of angle $2\pi/n$ and order n .
2. It contains a reflection S across a line ℓ through the origin.

Let's see the interplay between these:

- (c) Show that $SR = R^{-1}S$.
- (d) Show that $R^kS = SR^{-k}$ for all $k \in \mathbb{Z}$.

This shows that the elements of a dihedral group D_n breaks up into two types: powers of R (which can be written as R^k for some $0 \leq k \leq n-1$), and powers of R times S (so R^kS for some $0 \leq k \leq n-1$).

In particular, we conclude that the number of elements of D_n is $2n$.

- (e) What is the order of an element of the form R^kS ?



- 11 Draw a figure whose symmetry group consists of exactly 5 elements, and another one whose symmetry group consists of 6 elements.

- 12 The [Platonic Solids](#) are polyhedra showing a lot of symmetry. In dimension 3, a rotation is (almost) always about a line.

- (a) There is one rotation for which there exist no definitive rotation axis. Describe this 'rotation'.
- (b) Describe all rotations in the symmetry group of the cube. How many are there?

- 13 Deduce that the symmetry group of a regular tetrahedron consists of 24 transformations.

Tessellations

EXERCISES

- 14 A tessellation of the plane by using only one regular polygon $\{n\}$ is possible only in case $n = 3, 4$ or 6 . Why?

Extra exercises

EXERCISES

- 15 The only isometries in dimension 2 are rotations about a point, reflections in lines, translations over vectors or combinations of these. This can be proven by stating that an isometry is completely determined by the images of three non-collinear¹ points. Why is that, and why three and why non-collinear?
- 16 Find logos of companies with symmetry group D_3 , D_4 and D_6 .
- 17 Three-D print a [kuhkubus](#).
- 18 Download the 'KaleidoPaint' app by [Jeff Weeks](#) (available for Android and iOS). Make a tessellation of the plane with a symmetry of your choice.

¹collinear means: lying on one line.

Hints

- 1 Use $\mathbf{0} + \mathbf{0} = \mathbf{0}$.
- 2 (b) For which vector $\mathbf{b}$ does $T_{\mathbf{b}}\mathbf{p} = \mathbf{p}$ hold?
(c) For which vector $\mathbf{c}$ does $T_{\mathbf{b}}T_{\mathbf{c}} = I$ hold? Use (a) and (b).
- 5 (b) Check what G does with the standard base vectors $\mathbf{e}_1$ and $\mathbf{e}_2$.
- 7 (a) Use a proof by contradiction. What happens if $F^n \neq I$?
- 9 (c) Use $F^m F^n = F^{m+n}$.
(d) Use $F^{-1} = F^{-n}$.
- 10 See also the tutorial of week 1.
Note that a rotation over an angle θ_1 and a rotation over an angle θ_2 are considered to be the same if $\theta_1 - \theta_2$ is an integer multiple of 2π .
- 12 (a) For this rotation *all* lines are rotation axes.
(b) Think carefully about where the rotation axes should go with respect to the vertices, edges and faces of the cube. The angles are not all multiples of 90° !
- 14 Think of the angles at the vertices of the polygons, and remember that the tessellation must cover the whole plane without gaps.